

Michael Farquharson

905-616-8182 | mfarquharson1064@gmail.com | <https://www.michael-farquharson.com/>

EDUCATION

McMaster University

Hamilton, ON

Bachelor of Mechanical Engineering and Management, Graduated

Sept. 2015 – April 2021

- Achieved a cumulative GPA of 3.6/4.0, graduating Summa Cum Laude
- Specialized cross-facility program, combines a full mechanical engineering degree and the core elements of a commerce degree, with a specialization in entrepreneurship

McGill University

Montreal, QC

Masters of Science in Mechanical Engineering

Sept. 2021 – August 2025

- Conducted research as a part of the Applied Dynamics Group, a member of the Centre for Intelligent Machines
- Research focused on creating real time dynamic simulations with contact for use in teleoperation tasks
- Research was done in partnership with the National Research Council of Canada, Aerospace Manufacturing and Technology Centre (NRC-AMTC)

EXPERIENCE

HARP Technical Lead, Robotics Software Specialist

March 2025 – Present

Haply Robotics

Montreal, QC

- Robotics Software Specialist developing a rust-based concurrent robot control interface for a force feedback input device aimed at non-technical users called H.A.R.P. or the Haply Advanced Robotics Platform.
- Lead a small technical team of three, made up of software developers and robotics control engineers to develop a cutting edge tele-operation robotics control platform.
- Work with senior software architects to plan out and design HARP platform from the ground up to support high speed communication and industry level robustness and safety controls, while also being highly adaptable and modular to support future growth.
- Lead feature planning sessions with technical team using agile frameworks, deciding on priority levels, timelines, and implementation order to meet both external and internal requirements.
- Provide technical support to marketing and sales teams by preparing demos and addressing technical client inquiries.
- **Skills:** Project Management, Software Design, Control Systems Design, Concurrent Programming, Industrial Robotics, Mobile Robotics

Robotics Control Specialist

Dec. 2024 – March 2025

Haply Robotics

Montreal, QC

- R&D specialist developing a custom concurrent robot control interface for a force feedback input device aimed at non-technical users.
- Provide technical support to marketing and sales teams by preparing demos and addressing technical client inquiries.
- Maintain fast turnaround times on highly complex cross-dependent custom mechatronics projects.
- **Skills:** Industrial Robotics, Mobile Robotics, Control Systems Design, Concurrent Programming, Software Design

Mechatronics R&D

Dec. 2023 – Dec. 2024

Haply Robotics

Montreal, QC

- Rapidly prototype cutting-edge interface devices using various 3D printing technologies in an in-house fabrication workshop.
- Provide technical support to marketing and sales teams by preparing demos and addressing technical client inquiries.
- Maintain fast turnaround times on highly complex cross-dependent custom mechatronics projects.
- **Skills:** Rapid Prototyping, Control Systems Design, Concurrent Programming

Graduate Student Researcher

Sept. 2021 – August 2025

McGill University

Montreal, QC

- Conducted research in teleoperation of remote aerial vehicles in unstructured environments.
- Built drone simulation including contact physics from scratch.

- Developed ROS2-based test platform to ensure a low-latency, high-performance experience.
- Developed novel haptic control algorithm to provide high-performance human-machine interface.

Teaching Assistant

Sept. 2020 – April 2021

McMaster University

Hamilton, ON

- Facilitated the Integrated Cornerstone Design Projects (Engineering 1P13), guiding students through technical writing, design feasibility, and project development.
- Provided feedback on group projects, including material selection for wind turbine blades and the design of assistive devices.

Data Analyst Intern

May 2019 – Oct. 2019

Bombardier

Toronto, ON

- Solved complex data manipulation challenges independently, drastically reducing task time.
- Developed Python scripts for data cleaning, sorting, and analysis.
- Delivered quarterly presentations to senior management, contributing to significant departmental time savings.

Teaching Assistant

Jan. 2019 – April 2019

DeGroote School of Business, McMaster University

Hamilton, ON

- Assisted in delivering Innovation 1X03 under Professor Kenneth Owen.
- Evaluated written reflections and brainstorming activities related to innovation and entrepreneurship.
- Provided feedback on group research projects focused on initiating a theoretical business.

Undergraduate Summer Student

May 2018 – Sept. 2018

TMRL, McMaster University

- Awarded the Heater Fellowship to conduct research in waste heat recovery under Dr. James Cotton.
- Developed an analytical thermal system model using Modelica and Excel.
- Provided weekly progress updates and managed multiple projects under tight deadlines.

MAC Formula Electric

Sept. 2017 – Sept. 2018

McMaster University

- As Drivetrain Lead, spearheaded the design and manufacture of the drive train system for the race car, managed team workflow, and trained new members on necessary technical skills.
- As Team Member, assisted in the design and manufacturing of various car components.

Venture Engineering and Science Instructor

May 2017 – Aug. 2017

McMaster University

Toronto, ON

- Taught science and engineering concepts to campers (ages 10–12) through hands-on labs and projects.

Warehouse Worker

May 2016 – Aug. 2016

Bbq Galore

Toronto, ON

- Assembled and delivered BBQ units to customers, ensuring accurate and timely service.

Town of Oakville

Jul. 2015 – Aug. 2015 / Sept. 2014 – Jun. 2015

Oakville, ON

- Swim Instructor: Taught children aged 3 to 17 swimming techniques and safety.
- Lifeguard: Supervised pool patrons at QEPCCC pool to ensure safety.

PROJECTS

3D Printer | CAD, G-Code, C++, Soldering

Aug. 2017 – Aug. 2020

- Heavily modified a stock 3d printer to improve reliability and quality of prints
- Reinforced sections of the printer suffering from intense vibrations
- Installed auto-bed leveling sensor and software, plus an improved z-axis mechanism

Arduino Mechanical Clock | CAD, 3D Printing, Laser Cutting, Soldering, C++

May – Aug. 2020

- Instead of the two hands of the clock moving, a hour face and a minute face were moved
- Arduino powered stepper motors attached to 3d printed gears were used to move the faces

TECHNICAL SKILLS

Software Packages: vsCode, Git, Fork, Windows, Linux, ROS 2, Onshape, SolidWorks, Autodesk Inventor, FreeCad, Microsoft Office

Languages: Rust, Python, C/C++, TypeScript, Bash, MATLAB, Java, JavaScript, HTML/CSS, G-Code, SQL

Tools: FDM & SLA & SLS 3D Printers, Soldering Iron, Various Woodworking Tools, Milling Machine, Lathe, Drill Press

Robotics & Automation: Industrial Robotics, Mobile Robotics, Robot Operating System 2 (ROS2), Control Systems Design, Rapid Prototyping

Research & Development: Software Design, Teleoperation, Machine Learning, Classical Control Techniques, Remote Aerial Vehicles, Waste Heat Recovery

Teaching & Leadership: Project Facilitation, Team Management, Technical Support, Public Speaking, Curriculum Development

Additional Skills: Communication, Problem Solving, Time Management, Critical Thinking